FALCONS · EE308FZ 软件工程 · 作业四

SPORTS BUDDY

界面原型设计汇报

Interface Prototype Design · Sports Buddy

约得到 · 约得好 · 约得准

Falcons（第12组）· 12 屏交互原型（墨刀）



Prototype at a glance

12

interactive screens

login to review

5

SRS modules covered

M1–M5

108

real-browser checks

plus 188 Node checks

21

end-to-end steps

login → review

Tool and demo

Design tool: MoDao (墨刀), a specialised prototyping tool

Live demo: the 12 screens are embedded as interactive HTML pages, so the flow can be opened and clicked through during the defence

Links: modao.cc/proto/Ebn0JeMtntbdoIJ4M1YfJ · source: assignment4/prototype

Goal: an interactive prototype covering the full user journey, from login to review

Live demo: try the prototype



Scan to open the prototype

Interactive prototype (MoDao)

Scan the QR code with a phone, or click the link below to open the 12-screen prototype in device view.

Prototype

modao.cc/proto/Ebn0JeMtntbdoIJ4M1YfJ

Project repository (GitHub)

github.com/Falcons-FZU-MU-SE-GROUP12/SportsBuddy

Suggested demo path

Login → Discovery → Activity list → Detail → Sign up → Check-in → Review

Design concepts: buddy up easily, well, precisely

Easily

约得到

Design concept

One-sentence search returns ranked activities, so finding a game takes seconds.

Where to see it

P-02 Discovery Home · P-03 Activity List

Well

约得好

Design concept

Check-in, mutual reviews and a credit score make reliable people stand out and reduce no-shows.

Where to see it

P-07 Check-in · P-08 Manage · P-12 Review

Precisely

约得准

Design concept

Matching weighs sport, campus / time, skill level, distance and organizer reliability, and explains why.

Where to see it

P-02 Discovery Home (match reasons)

The prototype follows the product vision: “buddy up, easily, well and precisely”

Visual design principles

Mobile-first

A 390 × 844 phone frame is the main canvas; the layout also adapts to 320, 768 and 1280 px with no horizontal overflow.

Warm, energetic palette

Orange is the action colour, dark text sits on a light background, and the hierarchy stays clear.

Low cognitive load

One primary action per screen, clear status (full / joined / checked in) and explicit feedback after every action.



P-01 Login / Register



P-03 Activity List

Design process: six steps

1 Requirement mapping

The 12 screens were derived from the SRS (M1–M5) and the 41 use cases.

2 Information architecture

Login → discovery → list → detail → create → check-in → manage → my activities → me → profile → reviews.

3 Screen production

Each screen was built by its owner, then reviewed against the requirement it serves.

4 Interaction wiring

Buttons, tabs, forms, modals and page transitions were linked into one continuous flow.

5 Business-rule states

Full capacity, credit below 60, late cancellation, check-in code / location and permissions became visible states.

6 Review and regression

The testing group walked through the flow against the SRS; defects were fixed and re-verified.

Interface overview (1/2): from login to creating a game



P-01 Login / Register

Entry, demo account



P-02 Discovery Home

One-sentence search, recommendations



P-03 Activity List

Filters: sport, time, level, campus



P-04 Activity Detail (participant)

Join, see participants, status



P-05 Activity Detail (organizer)

Manage, see sign-ups



P-06 Create Activity

Time, place, headcount, level

Main flow: Login → Discovery → Activity list → Detail → Sign up → Check-in → Review

Interface overview (2/2): check-in, profile and reviews



P-07 Check-in

Code or location (within 300 m)



P-08 Manage Activity

Attendance, cancellation



P-09 My Activities

Organized / joined



P-10 Me

Profile, credit score, settings



P-11 Profile (other user)

Credit and reviews



P-12 Review Partners

Stars and impression tags

Together the 12 screens cover the whole journey, from login to review

Business rules made visible



P-04 Detail

join / cancel rules



P-07 Check-in

code or location



P-08 Manage

attendance

Rules shown as states

Capacity: status turns “Full” automatically

Credit < 60: cannot join new activities

Late cancel: less than 2 h before start: −5 pts

No-show: unticked at attendance: −15 pts

Check-in: 4-digit code, or location within 300 m

Permissions: only the organizer sees the check-in code

Requirements alignment: M1–M5 to screens

M1	User & authentication	P-01, P-10
M2	Activity publishing	P-06, P-05
M3	Matching & discovery	P-02, P-03
M4	Sign-up & participation	P-04, P-07, P-08, P-09
M5	Reliability & credit	P-04, P-11, P-12

All five SRS modules map to at least one screen

Results and verification

188 Node checks

automated regression suite

108 real-browser checks

titles, rendering, undefined / NaN, overflow, console errors

21 end-to-end steps

login → discover → sign up → check in → review

- ✓ All 12 screens reachable and clickable
- ✓ Main flow works end to end
- ✓ Business-rule states visible
- ✓ Responsive at 320 / 768 / 1280 px, no horizontal overflow
- ✓ Consistent colour, typography and components
- ✓ Real-browser verification and end-to-end flow passed
- ✓ Non-posed team discussion photo
- ✓ Public prototype link (MoDao)

Difficulties and solutions (1/2)

1 Layout collapsed: CSS class mismatch

Difficulty CSS used .proto-screen but the HTML used screen, so display: none never applied and all 12 screens rendered at once.

Fix Traced the selector mismatch and unified the class name to .screen.

Lesson Verify the rendered layout, not just the source; keep one naming convention.

2 Mobile top bar covered the screens

Difficulty The phone media query came before the base .proto-chrome rule and was overridden; the bar took about 130 px at 390 px.

Fix Moved the media query to the end of the stylesheet.

Lesson Cascade order matters; test responsive rules at real device widths.

3 Floating button on every screen

Difficulty The floating action button used position: fixed, so it hovered over all 12 screens.

Fix Scoped it under .screen.is-active so only the active screen shows it.

Lesson Scope global positioning when several screens share one page.

All three layout issues were solved and re-verified in the rendered prototype

Difficulties and solutions (2/2) and gains

4 Feedback messages and route parameters lost

Difficulty The router cleared messages before rendering, the login token was not persisted (401) and route parameters were dropped (/activities/NaN).

Fix Added a flashPending flag, stored the token on login and fixed route parsing to pass activityId / userId.

Lesson State that survives navigation needs an explicit lifecycle.

5 Class diagram overflowed on small screens

Difficulty The page overflowed by 12 px at 390 px because a grid column had a 360 px minimum.

Fix Switched to minmax(min(360px, 100%), 1fr) and regenerated; SVG diff was zero.

Lesson Test layout at the narrowest supported width.

Gains

Automated regression

188 Node checks and 108 real-browser checks (DevTools Protocol, headless Edge).

End-to-end journey test

21 steps: login → discover → sign up → check in → review.

Root-cause thinking

Fix causes, not symptoms; keep prototype, web app and share package generated from one source.

Team plan and progress



Preparation

10/26 – 10/27

Project set-up, screen ownership, flow map

Screen production

10/27 – 10/28

12 interactive screens, business-rule states

Consistency & review

10/28 – 10/29

Unified style, defect list, fixes

Publication

10/29 – 10/31

Public prototype link, testing report

Report & submission

10/31 – 11/01

Blog, PPT, final submission

Next steps before the defence

1. Finalise the presentation
2. Confirm the public MoDao link opens and navigates correctly

A one-week cycle; specific dates follow the course arrangement.

Roles and workload: 13 members, about 7.7% each

13

members · 1 team lead

6

functional groups

7.7%

Estimated share per member, 100% in total; adjusted by performance assessment

Project Mgmt — 1

Huang Zhiyuan (Team Lead)

Schedule, coordination, blog, integration, submission

Product Req. — 2

Zhang Zile · Ye Yifan

Prototype set-up, P-01–P-06, flow map, interaction details

Frontend — 4

Lin Zijun · Li Zhenhao · Li Haoze · He Jinghao

P-07–P-12, visual consistency, PPT layout

Backend — 4

Zhang Jingtian · Yu Chenghao · Zhang Xinyu · Xu Boyu

Business-rule states: capacity, credit, cancel penalty, check-in, permissions

Algorithm — 1

Cai Yaohua

Matching and reliability display, LLM research

Testing — 1

Wang Shaojie

Walkthrough, defect list, acceptance

Team collaboration

Repository and folders

All deliverables live in the shared GitHub repository under fixed folders: assignment4/prototype, slides, evidence and testing.

Work log

Every change is recorded in WORKLOG.md (date, member, type, files, summary, assignment), so contributions are traceable.

Task allocation and communication

Each screen and document has one owner and one reviewer; the lead tracks the critical path; daily coordination in the team chat.

Review and integration

The testing group walked through the prototype against the SRS; owners fixed and re-verified each defect; the share package was regenerated from one source.

Workload shares start as an equal estimate, then are adjusted by the team performance assessment

Team photo: reviewing the prototype together



Figure 15. Non-posed photo of the team discussing the prototype.



Figure 16. The team reviewing the prototype flow together.

Team feelings

“Clear scope and a working prototype beat a perfect plan: when problems appeared, we discussed them openly, fixed the root cause and re-verified the result.”

Huang Zhiyuan
Project Management (Team Lead)

“Whenever we face challenges, we communicate and discuss them together, each doing our part and collaborating seamlessly.”

Zhang Zile
Product Requirements

“From choosing Modao for prototyping to setting up the overall framework, everything feels efficient and gives me a lot of confidence for development.”

Yu Chenghao
Backend Development

“This team has shown me the power of teamwork. We split tasks clearly, and we always pull together to get through every difficulty.”

Wang Shaojie
Testing

Clear division of labour, GitHub work-log tracking and a test-driven review loop



谢谢大家

Q & A

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墨刀交互原型 · 源码：github.com/Falcons-FZU-MU-SE-GROUP12/SportsBuddy